

NETFLIX USER ANALYTICS

END-TO-END DATA ANALYTICS & BUSINESS INTELLIGENCE PROJECT

SQL • Python • Power BI • Excel

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1. Executive Summary & Project Overview

Netflix User Analytics is a comprehensive data analytics portfolio project designed to evaluate user streaming behavior, subscription financial metrics, content popularity, engagement trends, device distribution, and network infrastructure. The project transforms raw, AI-generated synthetic data into strategic business insights using an enterprise analytics stack.

20,000	3,239	\$21.34 M	\$6,587.27	3.83 / 5.0
TOTAL RECORDS	UNIQUE USERS	TOTAL REVENUE	AVG REVENUE / USER	AVG RATING

2. Technical Stack & Implementation Architecture

Tool / Tech	Role & Analytical Focus	Key Deliverables / Operations
MySQL 8.0	Relational Database, Query Engine, Data Modeling	Complex joins, CTEs, Window functions, Views, Stored Procedures, UDFs, B-Tree Indexes
Python 3.x	Data Cleaning, Preprocessing, EDA, Statistical Analysis	Pandas dataframe transformations, NumPy metrics, Matplotlib/Seaborn visualization suite
Power BI	Interactive BI Dashboard, DAX Modeling, Storytelling	Executive dashboards, dynamic slicing, drill-through pages, cross-filtering visuals
Microsoft Excel	Data Generation, Structuring & Initial Auditing	Synthetic dataset generation (39 attributes, 20,000 rows), schema formatting
GitHub	Version Control, Code Repository & Portfolio Host	Structured directory hosting SQL scripts, Jupyter notebooks, PBIX, and documentation

3. Dataset Schema & Structure (20,000 Records | July 2025 – July 2026)

The dataset spans **39 attributes** categorized into 10 operational domains. Note: This dataset is synthetic and AI-generated for analytical demonstration purposes.

Domain	Key Attributes Included	Analytical Scope
■ User Data	User ID, Age, Gender, Signup Date, Churn Status	Demographic breakdown & user lifecycle analysis
■ Subscriptions	Plan Type (Basic, Standard, Premium), Monthly Fee	Plan distribution & tier transition analysis
■ Revenue	Payment Method, Total Spend, Subscription Revenue	ARPU calculation & payment gateway performance
■ Content & Rating	Title, Genre, Content Type, Age Rating, User Rating	Genre popularity, rating distribution & title performance
■ Tech & Network	Device Type, OS, Internet Connection Type	Hardware ecosystem & streaming infrastructure load
■ Geography	Country, City	Regional revenue density & active user distribution

4. Relational Database & SQL Business Intelligence

The SQL component forms the backend foundation of the analytics workflow. Stored in MySQL, the relational schema implements sophisticated querying techniques across 21 business questions, utilizing Views, Stored Procedures, UDFs, and B-Tree Indexes.

SQL Script File	Core Purpose & Implementation Details
Database.sql	Database creation, table schema definition, primary/foreign key constraints.
Business_Queries.sql	21 complex analytical queries addressing revenue, engagement, churn, and ranking.
Views.sql	Reusable database views for high-level KPIs and dashboard connectivity.
Stored_Procedures.sql	Automated procedures for dynamic user segment reporting and country summaries.
Functions.sql	User-defined functions (UDFs) for custom metrics (e.g., ARPU calculation, churn-risk categorizer).
Indexes.sql	B-Tree index optimization on high-frequency filter columns (User_ID, Country, Genre, Watch_Date).

Key Business Queries & Advanced SQL Techniques

A. Advanced Window Functions: Ranking Top 3 Users per Country

```
WITH RankedUsers AS (  
    SELECT country, user_id, SUM(watch_time_minutes) AS total_watch_time,  
           DENSE_RANK() OVER (PARTITION BY country ORDER BY SUM(watch_time_minutes) DESC) as rnk  
    FROM netflix_user_analytics  
    GROUP BY country, user_id  
)  
SELECT country, user_id, total_watch_time  
FROM RankedUsers WHERE rnk <= 3 ORDER BY country, rnk;
```

B. Running Cumulative Revenue & Monthly Engagement Trends

```
SELECT watch_date, SUM(subscription_fee) AS daily_revenue,  
       SUM(SUM(subscription_fee)) OVER (ORDER BY watch_date) AS running_total_revenue,  
       AVG(completion_percentage) OVER (ORDER BY watch_date ROWS BETWEEN 6 PRECEDING AND CURRENT ROW) AS rolling_avg_completion  
FROM netflix_user_analytics GROUP BY watch_date ORDER BY watch_date;
```

Summary of Core Business Questions Answered

Category	Key Business Question Addressed	SQL Techniques Applied
Revenue	Which subscription tier yields highest total revenue & ARPU?	GROUP BY, Aggregate SUM/AVG, CASE
Content	Top 10 most-watched titles & genre watch time distribution?	ORDER BY DESC, LIMIT, Aggregate joins
Engagement	Average completion percentage by genre & age rating?	Nested aggregations, conditional filtering
User Behavior	User categorization based on watch time & churn risk?	CASE WHEN logic, CTEs, Windowing
Infrastructure	Device, OS, and internet connection share per region?	Multi-column aggregation, subqueries

5. Python Exploratory Data Analysis & Revenue Insights

Python (Pandas, NumPy, Matplotlib, Seaborn) was utilized to execute rigorous data cleaning, missing value handle, type standardization, and deep exploratory data analysis (EDA). The notebook Netflix_User_Analytics.ipynb structures the analytics pipeline from raw data to published KPIs.

Data Cleaning & Preprocessing Pipeline

- **Date Normalization:** Converted Watch_Date and Signup_Date to datetime objects; handled timezone offsets.
- **Type Casting & Formatting:** Cleaned currency symbols from subscription fees; ensured watch times were standard numeric integers.
- **Missing Value & Duplicate Audit:** Confirmed zero nulls across primary key paths; removed 0 duplicate session entries.
- **Outlier Analysis:** Evaluated extreme watch-time values using IQR (Interquartile Range) to confirm synthetic integrity.

Financial & Subscription Tier Breakdown

Subscription Plan Tier	Total Revenue (\$)	Revenue Share (%)	ARPU (\$)	Strategic Tier Insights
Premium Tier	\$15,422,400.40	72.28%	\$8,125.61	Primary revenue engine; strong retention among heavy streaming users.
Standard Tier	\$4,120,550.20	19.31%	\$5,210.12	Mid-tier staple; high conversion potential to Premium via features.
Basic Tier	\$1,793,228.86	8.41%	\$3,145.50	Entry-level plan; serves cost-sensitive regional markets.
Total / Portfolio Avg	\$21,336,179.46	100.00%	\$6,587.27	Solid monetization foundation across 3,239 users.

Geographic Performance & Market Density

Revenue generation is highly concentrated across key global markets, with Asian and North American markets leading overall financial output.

Top Country	Active Users	User Share (%)	Key Revenue-Contributing Cities
United States	711	21.95%	New York, Los Angeles, Chicago, Houston
India	656	20.25%	Delhi, Kolkata, Mumbai, Bengaluru
United Kingdom	319	9.85%	London, Manchester, Birmingham
Canada	271	8.37%	Toronto, Vancouver, Montreal
France	248	7.66%	Paris, Lyon, Marseille
Germany	240	7.41%	Berlin, Munich, Hamburg
Japan	223	6.88%	Tokyo, Osaka, Sapporo
South Korea	197	6.08%	Seoul, Busan, Incheon, Daegu

6. Content Popularity, User Engagement & Network Analytics

Content & Genre Performance Breakdown

Rank	Genre	Avg Rating
1	Mystery	3.88
2	Thriller	3.87
3	Biography	3.86
4	Action	3.86
5	Documentary	3.85
6	Animation	3.84
7	Adventure	3.84
8	Fantasy	3.84
9	Drama	3.83
10	Horror	3.83
11	Romance	3.82
12	Comedy	3.82
13	Family	3.82
14	Sci-Fi	3.81
15	Crime	3.79

Rank	Top Most-Watched Title	Views
1	Ancient Ascension	432
2	House of Obsidian	413
3	Sacred Ascension: Redemption	410
4	Quiet Skyline: Redemption	405
5	Sons of Ember	395
6	Sons of Ravens	392
7	Quiet Requiem Reborn	390
8	Shattered Legacy Awakens	388
9	Daughters of Wolves	385
10	Broken Paradox: Redemption	382

Top 10 Highly Engaged Power Users (By Total Watch Time)

User ID	Watch Time (min)	User ID	Watch Time (min)
USR101349	2,822 min	USR102042	2,260 min
USR102171	2,637 min	USR102720	2,176 min
USR101133	2,546 min	USR103063	2,161 min
USR100861	2,539 min	USR103346	2,116 min
USR101510	2,401 min	USR103411	2,103 min

Infrastructure & Network Delivery Breakdown

Connection Type	Streaming Sessions	Session Share (%)	Primary Hardware Ecosystem
Wi-Fi Network	8,184	40.92%	Mobile, Tablets, Laptops (Domestic / Portable)
Fixed Broadband	5,934	29.67%	Smart TV, Desktop, Gaming Consoles (Living Room)
Mobile Data (4G/5G)	5,882	29.41%	Smartphones, On-the-go streaming

7. Power BI Dashboard Architecture & Business Recommendations

Power BI Business Intelligence Dashboard Overview

The Power BI dashboard (Netflix Dashboard.pbix) provides executive decision-makers with interactive data visualization, dynamically linked slicers, dynamic cross-filtering, and drill-through analysis across all 20,000 streaming records.

Dashboard Visual Component	Interactive Functionality & Analytical Purpose
Executive KPI Card Cluster	Displays real-time Total Revenue, Active Users, ARPU, Total Watch Time, and Rating.
Revenue Tier & Geo Maps	Decomposes subscription revenue by Plan Type and interactive global choropleth map.
Monthly Viewing Trendline	Tracks seasonality and watch-time peaks from July 2025 to July 2026.
Device & Network Matrix	Visualizes streaming share across Mobile, Smart TV, Laptop, Tablet, Desktop, and Consoles.
Drill-Through Genre Page	Allows deep-dive exploration into individual title ratings, age ratings, and completion rates.

Strategic Business Recommendations

1. Tier Upselling Strategy	Convert Basic (8.4% revenue) and Standard (19.3%) users to Premium through targeted trial offers, emphasis on 4K multi-device streaming, and exclusive feature bundles.
2. Content Investment Focus	Allocate production and licensing capital toward Fantasy, Action, and Mystery genres, which yield the highest cumulative watch times and top user ratings (3.88 avg).
3. Power User Retention	Implement loyalty initiatives and early-access preview perks for top decile users (>2,000 min watch time) to prevent churn among highest-value subscribers.
4. Network & Device Optimization	Optimize encoding algorithms specifically for Mobile Data (29.4% share) and Wi-Fi networks to reduce buffering and improve video completion rates.

Project Execution & Author Details

Author: Ratnesh Chauhan Role Focus: Aspiring Data Analyst Business Analyst Power BI Developer GitHub Portfolio: github.com/Ratnesh8577 LinkedIn Profile: linkedin.com/in/ratnesh-chauhan-41a113279/ LeetCode: leetcode.com/u/RatneshChauhan279/	Project Outcome: Demonstrated end-to-end data pipeline execution from synthetic Excel data generation through MySQL database warehousing, Python statistical EDA, to executive Power BI interactive storytelling.
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